Mohamed Ahmed Ali

Backend Software Engineer | NestJS • TypeScript • Node.js • PostgreSQL
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LINKS

LinkedIn: linkedin.com/in/mohamed-ahmed-ali-a019a9218
GitHub: github.com/Mali-98 | Portfolio: mali-98.github.io/Portfolio

PROFESSIONAL SUMMARY

Backend Software Engineer with 5 years of experience building scalable RESTful APIs, role-based access control (RBAC), and authentication with NestJS, TypeScript, PostgreSQL, and Firebase. Delivered AI chatbot, e-learning, healthcare, and e-commerce platforms. Background in SQL data analysis, integration testing, and embedded systems.

WORK EXPERIENCE

Sep 2025 – Present

Senior Software Developer, BotBat — United Arab Emirates

- **Built a full-stack e-commerce chatbot platform** where an AI assistant analyzes user queries, retrieves relevant inventory items, and guides users to purchase.
- **Designed a node-based workflow engine** that routes conversations between AI agents and human support agents based on user intent and context.

Jul 2025 – Mar 2026

Backend Developer, Imdova — Cairo

- **Developed the backend** for a Courses platform (on-demand and live training) for undergraduate medical students.
- **Built scalable RESTful APIs** covering 5 domains: user accounts, course enrollment, video content, assessments, and certification.

Apr 2025 – Jun 2025

Backend Developer, Spotter — Cairo

- **Contributed to a fitness coaching platform** delivering 3 core features: personalized advice, progress tracking, and AI chatbot interactions.
- **Ensured clean architecture** through modular service design and scalable authentication strategies.

Apr 2023 – Jul 2025

Backend Developer, Arcon Corporation — Cairo

- **Designed and developed scalable RESTful APIs** for educational and healthcare platforms, including a multi-tenant LMS where each customer organization manages unlimited courses with 51+ experiments per course.
- **Built enrollment and progress tracking** that supports any number of students per course and records each student's score and attempt number per experiment.
- **Implemented RBAC and Firebase Authentication** for secure user and permission management.
- **Wrote integration tests** validating module interactions and business-logic requirements.

Sep 2021 – Mar 2023

Embedded Software & Robotics Engineer, Arcon Corporation — Cairo

- **Developed embedded systems for 51+ scientific projects** using Arduino, sensors, and electronic components.
- **Developed Siminds LabsForHome**, a 3D simulation platform with 51+ experiments in 2 languages (Arabic and English), eye-tracking and voice-command accessibility modes, and real-time sync with physical Arduino kits.
- **Led robotics development and supervised VEX robotics competitions**, mentoring teams and optimizing performance strategies.

SELECTED PROJECTS

botbat.io

Botbat – AI Omnichannel Customer Engagement Platform

- **Connected 7 channels** (WhatsApp, Instagram, Messenger, Telegram, email, SMS, and web chat) in one customer engagement platform.
- **Enabled email and WhatsApp campaigns to up to 5,000 users** per send, with a customer data platform (CDP) storing **10,000+ profiles** created automatically per recipient.
- **Delivered intelligent chatbots**, workflow automation, analytics, and a unified agent portal.

medicova.net

Medicova – Healthcare Professionals Platform

- **Delivered 4 services** for healthcare professionals in Egypt and the Gulf: job listings, accredited courses and diplomas, exam question banks, and a medical products shop.
- **Supported preparation for 2 licensing exams** (USMLE and NCLEX-RN) through exam question banks.

Demo video

Medyour – Medical Booking App

- **Enabled booking of 3 service types**: appointments, surgeries, and blood sampling operations.
- **Integrated a third-party payment system** for secure transactions.

Demo video

LMS System (NestJS, Firebase Authentication, PostgreSQL)

- **Developed a learning management backend** with RBAC for 3 roles (admin, teacher, student), covering courses, classes, and assignments.

Demo video

Real-Time Chat Application (NestJS, WebSockets, Vue.js)

- **Built WebSocket-based messaging** with instant updates and typing indicators across multiple users.

EDUCATION

Sep 2016 – Jul 2021

Bachelor of Science in Mechatronics Engineering, Faculty of Engineering, Ain Shams University

Cairo

Graduated with Very Good (GPA 2.98) | Graduation Project Grade: A+

SKILLS

Backend	NestJS, Node.js, TypeScript, JavaScript, PHP, Laravel, RESTful APIs, WebSockets, JWT, Firebase Authentication, RBAC
Databases	PostgreSQL, SQL, MongoDB, Google Firestore, BigQuery
Data & AI	Power BI, SQL Data Analysis, Python, Machine Learning, AI Chatbots
Testing & Practices	Integration Testing, Software Testing, Clean Architecture
Other	Java, C/C++, Vue.js, Arduino and Embedded Systems, UX Fundamentals

CERTIFICATIONS

Software Testing – ITI MaharaTech
IoT Development Using M.O.T Platform – ITI MaharaTech
Gamification in eLearning – ITI MaharaTech
UX Design Fundamentals – ITI MaharaTech